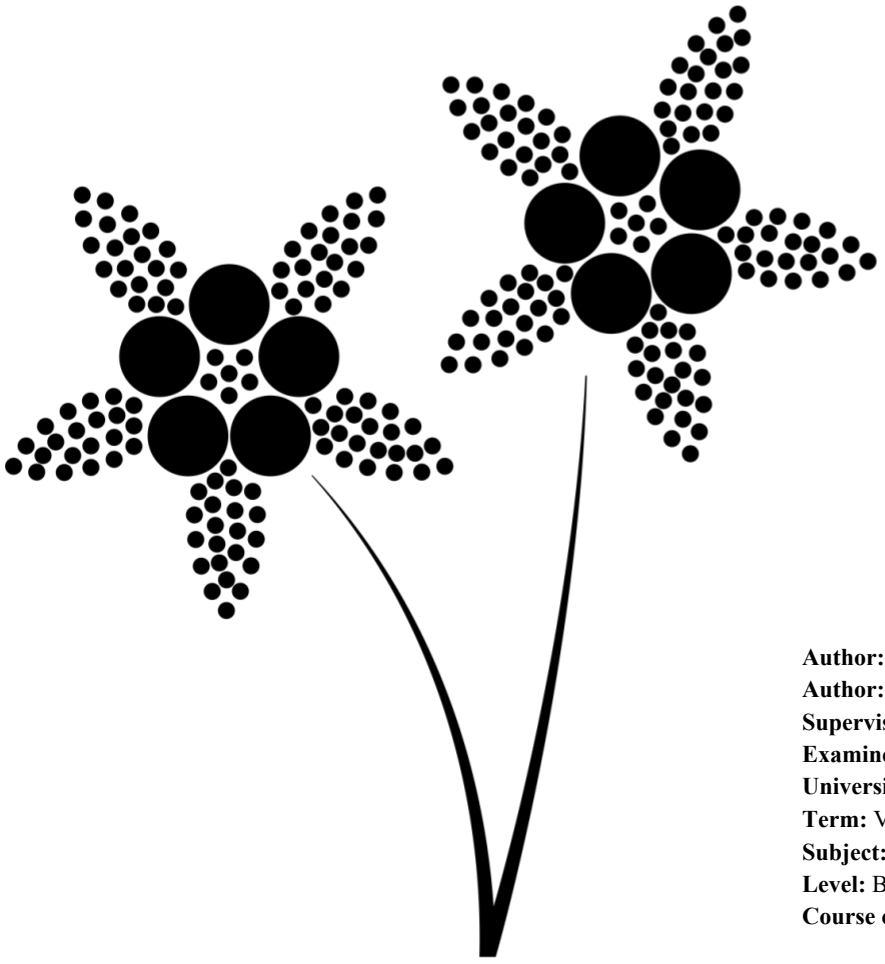
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Designing for Calm: Explorling intrusiveness in supportive interventions for digital work

A Research through Design study using design scenarios and interviews with IT professionals



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Abstract

When working at a computer for long periods of time, breaks may support well-being and recovery. At the same time, a notification or interruption may appear at the wrong time, making it difficult to return to the same level of focus and workflow as before. Even when the intention is to support the user, interruptions can therefore become stressful if they break concentration at inappropriate moments.

In our study, we wanted to explore how people working in digital environments with frequent interruptions experience different types of supportive interventions. The study explored where the boundary exists between supportive and overly intrusive interventions, and how different levels of intrusiveness affect focus, workflow continuity, and perceived control.

To explore this, we used a Research through Design approach, where we designed three scenarios representing different levels of intrusiveness: low, moderate and high intrusiveness. The low-intrusiveness scenario was inspired by principles of calm technology, where information remains in the periphery of attention without demanding immediate focus. In the more intrusive scenarios, the degree of interruption increased while user control decreased. We interviewed six IT professionals who were asked to imagine themselves working on a focused coding task when the interventions appeared.

The findings showed that high intrusiveness was experienced as highly disruptive. Participants described losing control over their workflow and found it difficult to return to the same level of concentration after the interruption. In contrast, low intrusiveness was generally perceived as more acceptable, as users retained control and could choose whether and when to engage with the notification.

The most notable findings emerged in the moderate level of intrusiveness, where participants' experiences varied considerably. The same type of intervention could be perceived as supportive by some participants and disruptive by others. Even when users could ignore or postpone the notification, the intervention still created attentional demands or introduced moments of decision-making during focused work.

For interaction design, these findings suggest that intrusiveness cannot be understood as a binary, nor as a matter of simply finding a middle ground. Instead, experiences of support and disruption appear to depend on factors such as timing, workflow, attentional state, perceived control and situational context. This highlights opportunities for future digital well-being systems and supportive technologies to become more adaptive and context-sensitive, calibrating interventions in ways that better align with users’ current needs and workflow patterns.

Keywords

Intrusiveness, calm technology, user experience, digital well-being, interaction design, workflow continuity, computer-based work, interruptions.

Sammanfattning

Att designa för lugn - En studie om intrusivitet i stödjande interventioner för digitalt arbete

När man arbetar vid en dator under längre perioder kan pauser bidra till välbefinnande och återhämtning. Samtidigt kan en notifikation eller ett avbrott komma vid fel tillfälle, vilket kan göra det svårt att återgå till samma nivå av fokus och arbetsflöde som tidigare. Även när avsikten är att stödja användaren kan avbrott därför bli stressande om de bryter koncentrationen vid olämpliga tillfällen.
I vår studie ville vi undersöka hur personer som arbetar i digitala miljöer med frekventa avbrott upplever olika typer av stödjande interventioner. Studien undersökte var gränsen går mellan stödjande och alltför påträngande interventioner, samt hur olika nivåer av intrusivitet påverkar fokus, kontinuitet i arbetsflödet och upplevd kontroll.
För att undersöka detta använde vi en Research through Design-ansats, där vi utformade tre scenarier som representerade olika nivåer av intrusivitet: låg, måttlig och hög. Scenariot med låg intrusivitet inspirerades av principer från Calm Technology, där information förblir i periferin av uppmärksamheten utan att kräva omedelbart fokus. I de mer intrusiva scenarierna ökade graden av avbrott samtidigt som användarens kontroll minskade. Vi intervjuade sex IT-professionella som ombads föreställa sig att de arbetade med en fokuserad programmeringsuppgift när interventionerna uppstod.
Resultaten visade att hög intrusivitet upplevdes som mycket störande. Deltagarna beskrev att de förlorade kontrollen över sitt arbetsflöde och hade svårt att återgå till samma koncentrationsnivå efter avbrottet. Låg intrusivitet upplevdes däremot generellt som mer acceptabel, eftersom användarna behöll kontrollen och själva kunde välja om och när de ville interagera med notifikationen.
De mest framträdande resultaten framkom vid den måttliga nivån av intrusivitet, där deltagarnas upplevelser varierade avsevärt. Samma typ av intervention kunde upplevas som stödjande av vissa deltagare och störande av andra. Även när användarna kunde ignorera eller skjuta upp notifikationen skapade interventionen fortfarande krav på uppmärksamhet eller introducerade stunder av beslutsfattande under fokuserat arbete.
För interaktionsdesign tyder dessa resultat på att intrusivitet varken kan förstås som något binärt eller som en fråga om att helt enkelt hitta en medelnivå. Upplevelsen av stöd och störning verkar i stället bero på faktorer som timing, arbetsflöde, uppmärksamhetstillstånd, upplevd kontroll och situationskontext. Detta synliggör möjligheter för framtida system för digitalt välbefinnande och stödjande tekniker att bli mer adaptiva och kontextkänsliga genom att kalibrera interventioner på sätt som bättre överensstämmer med användarnas aktuella behov och arbetsmönster.

Nyckelord

Intrusivitet, Calm Technology, användarupplevelse, digitalt välbefinnande, interaktionsdesign, kontinuitet i arbetsflödet, datorbaserat arbete, avbrott.

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This thesis was also a unique journey for us as co-authors, as we did not know each other before starting this project. Throughout the process, we learned how to collaborate, communicate, and support each other through both challenges and stressful moments. What began as an unfamiliar collaboration gradually developed into a meaningful teamwork experience that helped us grow both academically and personally.
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Table of Contents

1 Introduction 1
1.1 Background and problem statement 1
1.2 Previous research 2
1.3 The relevance and positioning of the research 3
1.4 Purpose and research questions 4
1.5 Delimitations 4
1.6 Thesis Structure 5

2 Literature Review and Theoretical Framework 6
2.1 Literature search strategy 6
2.2 Calm technology 7
2.3 Intrusiveness as a Design Dimension 7

3 Methodology 10
3.1 Research Approach 10
3.2 Research through Design (RtD) 10
3.2.1 Design process and Data collection 10
3.2.2 Phase 1 — Exploration 10
3.2.3 Phase 2 — Ideation and Concept Development 11
3.2.4 Phase 3 — Prototyping 11
3.2.5 Phase 4 — Evaluation with Users 12
3.3 Data analysis 12
3.4 Ethical Considerations 13

4 Results 14
4.1 Phase 1 — Exploration 14
4.2 Phase 2 — Ideation and concept development 14
4.3 Phase 3 — Prototyping 16
4.3.1 Scenario 1 — High Intrusiveness (full-screen notification) 16
4.3.2 Scenario 2 — Moderate Intrusiveness (partial notification with options) 18
4.3.3 Scenario 3 — Low Intrusiveness (subtle status indicator) 20
4.4 Phase 4 — Evaluation with users 21
4.4.1 Scenario 1 — High Intrusiveness (full-screen notification) 24
4.4.2 Scenario 2 — Moderate Intrusiveness (partial notification with options) 25
4.4.3 Scenario 3 — Low Intrusiveness (subtle status indicator) 25
4.5 Comparative findings 26
4.6 Design preferences 27

5 Analysis 28
5.1 Thematic Analysis Approach 28
5.2 Control 28
5.3 Disruption 29
5.4 Focus 30

6 Discussion 32
6.1 Answering the Research Question 32
6.2 Discussion of findings in relation to Research question 33
6.3 Contribution to Interaction design knowledge 34
6.4 Discussion of Method 35
6.5 Limitations 36

7 Conclusion 38
7.1 Future research 38
8 References..... 40

List of figures

Figure 1 — Visual interpretation of Scenario 1
Figure 2 — Visual interpretation of Scenario 2
Figure 3 — Visual interpretation of Scenario 3
Figure 4 — Comparative overview of experiences across the three levels of intrusiveness

List of tables

Table 1 — Explored interaction concepts and selection criteria
Table 2 — Overview of participants and selected interview quotes

Appendices

Appendix 1 Informed Consent form
Appendix 2 Interview Guide
Appendix 3 Overview of AI tools used in this thesis
Appendix 4 Division of work between writing partners

Introduction

Computer-based tools and communication platforms form the core work environment for many IT professionals today. Daily work often involves multiple applications, continuous communication, and parallel information flows, which place increasing demands on users' attention and ability to stay focused. As a result, questions concerning well-being, attention management, and workflow have become increasingly relevant within user experience (UX) and interaction design.

While digital well-being initiatives aim to support sustainable and balanced technology use, supportive features may themselves introduce additional attentional demands if not carefully designed. The question is therefore not only whether systems provide support, but how that support is experienced during focused work.

1.1 Background and problem statement

Research on digital well-being emphasises that maintaining focus should not be understood solely as an individual responsibility. Churchill (2020) argues that technological systems play a significant role in shaping users' cognitive load and attentional capacity. Features intended to support users, such as reminders, restrictions, or automated adjustments, may unintentionally increase interruption if poorly designed.

Weiser and Brown (1996) introduced the concept of Calm Technology, arguing that interactive systems should move between the centre and periphery of attention rather than constantly demanding focus. Building on this foundation, contemporary design research further emphasises low-intrusive interaction and peripheral awareness in interactive systems (Hassan et al., 2024). From this perspective, the positioning of interaction within users' attentional field becomes a key design consideration. Not all interactions require immediate attention. Systems can be designed to remain in the periphery of users' awareness and only move toward the centre of attention when necessary. Designing for peripheral awareness allows users to maintain focus on primary tasks while still remaining aware of relevant contextual changes. These ideas have also informed research on notification systems, particularly in understanding how digital interruptions affect users' attention and task performance.

Empirical research on notifications further demonstrates that system-initiated alerts can significantly influence ongoing work. Iqbal and Horvitz (2010) show that notifications can provide valuable awareness while also triggering task switching and cognitive disruption, depending on timing and salience. This suggests that intrusiveness is not a binary condition but exists along a spectrum shaped by design decisions. Understanding intrusiveness as a spectrum highlights the importance of how and when systems communicate with users. Factors such as timing, frequency, and presentation influence whether a notification supports awareness or disrupts workflow continuity. As a result,

designing effective notification systems requires careful consideration of how interactions enter the user's attentional space.

As artificial intelligence becomes more common in work-related systems, adaptive forms of support have also emerged. AI-based features can adjust their behaviour based on user context, activity, or previous interaction patterns. Jingar (2025) notes that while such systems create new opportunities for personalised support, they also introduce design challenges related to transparency, predictability, and user control. These challenges are particularly relevant when system behaviour changes automatically. Users may experience uncertainty or reduced agency, particularly if system actions are not clearly communicated. Adaptive systems therefore introduce additional complexity to the design of digital work environments. Because system behaviour may change automatically based on context or inferred user needs, system interventions may occur at moments that users did not explicitly initiate. These features aim to support productivity or well-being; they also raise design questions about how and when systems should intervene without becoming intrusive.

From a UX perspective, the central issue is therefore not the presence of adaptive functionality itself, but how different forms of system intervention are experienced during focused work. Interaction qualities such as intrusiveness, timing, and degree of control shape whether a feature feels supportive or disruptive (Iqbal & Horvitz, 2010; Czerwinski et al., 2004). When systems intervene or adapt without clear feedback or explanation, they risk being perceived as disruptive or controlling (Jingar, 2025). This creates a tension between the intention to support focus and the risk of interrupting the users' ongoing work, a challenge that has also been highlighted in calm interaction research (Weiser & Brown, 1996; Hassan et al., 2024).

Despite growing interest in adaptive support systems for well-being, limited attention has been given to intrusiveness as a deliberately designed and graded interaction variable in computer-based work contexts. There is insufficient understanding of how varying degrees of intrusiveness shape users' experience of focus and workflow in computer-based work environments. This suggests a need for a design exploration of how low-intrusive support can be shaped and evaluated from a UX perspective.

1.2 Previous research

Previous research relevant to this study can be understood across three interrelated strands.

First, research on digital well-being emphasises the responsibility of design in shaping user experience. Churchill (2020) critiques approach that place the burden of attention management on individuals and calls for more reflective design practices that consider cognitive and emotional impact.

Second, research on Calm Technology frames disruption as a design problem related to attentional demand. Rather than focusing on task performance alone, this strand of

research emphasises how interaction qualities shape users' experience of cognitive load and engagement (Weiser & Brown, 1996; Hassan et al., 2024). However, while this viewpoint highlights the importance of reducing attentional demand, it does not specify how graded levels of intrusiveness can be systematically structured and examined in focused work contexts.

Third, research on interruptions and task switching provides empirical insight into how system interventions affect workflow continuity. Diary studies of information workers show that interruptions significantly influence task resumption and perceived task difficulty. Czerwinski, Horvitz, and Wilhite (2004) found that tasks returned to after interruption were often longer, more complex, and more frequently disrupted. These findings suggest that interruptions have consequences not only for momentary attention but also for sustained workflow and cognitive effort. Similarly, Iqbal and Horvitz (2010) demonstrate that notification timing and salience shape perceived disruption and task switching, reinforcing the idea that system interventions can be differentiated according to their level of attention demand and interruption intensity.

Research on AI-enabled adaptive systems introduces further complexity. Jingar (2025) discusses how adaptive technologies may offer personalised support but also create tensions related to transparency and perceived control. However, much of this research focuses on system capability or organisational implications rather than on how graded levels of intrusiveness influence lived experience during focused work.

Despite these efforts, it remains unclear how different degrees of system interventions influence the experience of focused work. While prior research has shown that interruptions can disrupt workflow and that factors such as timing and notification salience influence perceived disruption, less attention has been paid to how graded levels of intrusiveness are experienced in focused work contexts from a user experience and interaction design perspective.

1.3 The relevance and positioning of the research

This study is positioned within the field of UX and interaction design, drawing on insights from interruption research and Calm Technology. Interruption research (Czerwinski et al., 2004; Iqbal & Horvitz, 2010) provides empirical insight into how system-initiated alerts influence workflow continuity and cognitive effort, while Calm Technology (Weiser & Brown, 1996; Hassan et al., 2024) offers a conceptual framework for understanding attention as distributed between focal and peripheral awareness. Together, these perspectives frame intrusiveness as an interaction quality shaped by the degree of system intervention and its impact on user perception.

Rather than treating interruption as an unavoidable side effect of system functionality, this study investigated how intrusiveness can be deliberately shaped and evaluated as part of interaction design.

The increasing integration of adaptive and AI-enabled features in contemporary work systems forms the contextual background of the study (Jingar, 2025). However, the focus remains on how varying degrees of system intervention are experienced during focused work, rather than on the technical implementation of adaptive systems.
By combining insights from interruption research, calm technology, and adaptive system design, the study seeks to bridge theoretical discussions of attention with practical design exploration. Intrusiveness is treated not as an inherent system property, but as a quality emerging from the relationship between system behaviour and user perception.
The primary contribution of this thesis takes the form of design knowledge: structured variations in system intervention that make different interaction qualities of intrusiveness concrete, examinable, and comparable from a user experience perspective. These design scenarios function as knowledge contributions in themselves; they embody claims about what low-intrusive interaction might look like in computer-based work that cannot be fully expressed in propositional form.

1.4 Purpose and research questions

The aim of this study is to explore how supportive features can be designed to help IT professionals maintain focus during computer-based work. The study examines how different levels of intrusiveness influence users' experience and sense of control in everyday work situations.
Main research question:
How can different levels of intrusiveness in supportive features shape IT professionals' experience of focus, workflow, and perceived control during everyday computer-based work?

1.5 Delimitations

This study focuses on computer-based work among IT professionals and explores levels of intrusiveness in supportive digital features during focused work activities. Therefore, the thesis is not intended to represent all types of digital workers or work environments. The focus on this user group allows the study to examine situations where focused digital work and system notifications are common.
The study uses conceptual design scenarios to explore how different degrees of intrusiveness in supportive digital features may influence users' experience of focus, interruptions, and perceived control. Instead of developing or evaluating a fully implemented software system, the intention is therefore not to produce a finished design solution, but to generate insights that may inform future design work and further research.
Finally, the study does not examine organizational differences, cultural factors, or objective productivity measures. It focuses on users' subjective experiences of interruptions and attention in digital work contexts. These aspects may influence how

interruptions are experienced, but they fall outside the scope of this exploratory design study

1.6 Thesis Structure

- Chapter 2 reviews literature on digital work interruptions, calm technology, and intrusiveness, establishing the theoretical framework for understanding system interventions in digital work environments.
- Chapter 3 describes the Research through Design methodology, including the design process, scenario development and qualitative data collection through semi-structured interviews.
- Chapter 4 presents the outcomes of the design process, including the three conceptual scenarios representing different levels of intrusiveness and the findings generated through the user interviews.
- Chapter 5 analyses the empirical material through a thematic analysis focusing on the themes of control, disruption and focus.
- Chapter 6 discusses the findings in relation to the research question, reflects on the study's contribution to interaction design knowledge and addresses methodological considerations and limitations.
- Chapter 7 concludes the thesis and outlines potential directions for future research and design work related to digital well-being and supportive interventions.

2 Literature Review and Theoretical Framework

2.1 Literature search strategy

The literature search aimed to identify research related to digital work interruptions, calm technology and intrusiveness in digital systems. The purpose was to build a theoretical understanding of how system interventions influence attention, workflow and perceived control in computer-based work.

Academic databases including ACM Digital Library, Scopus and Google Scholar were used to locate relevant articles within human-computer interaction and related fields. When searches produced a large number of technically oriented AI papers, Boolean operators (NOT AI) were used to narrow down the results towards studies focusing on user experience rather than system implementation.

The process began with broader exploratory searches focusing on digital well-being in digital work environments. Keywords such as digital well-being, workplace well-being and digital work environment were used to identify research discussing how digital technologies influence attention, work boundaries, and perceived well-being in digitally mediated work contexts.

As the research focus became clearer, the search gradually shifted towards studies on interruptions and notifications in digital work. Keywords such as interruptions, notification systems, attention management, and information overload were used to locate studies examining how system-initiated notifications affect workflow continuity and cognitive effort.

Additional searches explored design-oriented perspectives, including calm technology, peripheral interaction, and ambient information, to identify research discussing how digital systems can communicate information without constantly demanding the user's full attention.

The search developed iteratively throughout the research process. Relevant publications were identified through keyword searches as well as reviewing reference lists of previously identified articles. The final selection focused on peer-reviewed research within human-computer interaction and interaction design that addressed interruptions and user experience in computer-based work contexts.

Research on digital work, interruptions and attentional demand provides the theoretical context for understanding how system-initiated interventions influence ongoing work. Within HCI, several strands of research have examined how digital systems interrupt workflows, how notifications are experienced in everyday work and how attentional demand shapes the experience of digital interaction. Workflow in this context refers to task progression, continuity of thought, concentration, and momentum during ongoing work. Together, these perspectives help frame intrusiveness as a graded interaction quality rather than a binary condition.

The increasing digitalization of work has created environments characterized by continuous information flows and frequent digitally mediated work interruptions across different media types such as emails, instant messages, and calendar entries. Research in human-computer interaction (HCI) demonstrates that workers frequently switch between tasks during the workday and manage multiple activities in parallel. A diary study by Czerwinski et al. (2004) showed that complex and long-term tasks are particularly difficult to resume after interruptions. Returning to a suspended task often requires reconstructing context and remembering prior actions, which increases cognitive effort. Complementary experimental research further indicates that the impact of interruptions depends on their relevance to the ongoing task (Nadj et al., 2023). Interruptions perceived as relevant are easier to integrate, whereas irrelevant interruptions tend to be more disruptive and demand greater recovery effort. These findings suggest that interruptions are not inherently harmful; rather, these effects are shaped by context and task alignment. Churchill (2020) frames digital well-being as a matter of system design rather than individual responsibility. She argues that supportive features such as reminders or restrictions may unintentionally increase cognitive burden if they are poorly designed. Instead, she calls for context-aware systems capable of adapting interventions based on users' cognitive capacity and situational urgency. While this perspective highlights the importance of adaptive and supportive tools, it leaves open questions about how different degrees of intrusiveness are experienced in everyday work settings.

2.2 Calm technology

Calm Technology originates from Weiser and Brown's (1996) vision of ubiquitous computing, where technology should support human activity without dominating the attention. Rather than competing for cognitive resources, interactive systems are expected to move between the centre and the periphery of awareness. The core assumption is that human attention is limited and should be preserved rather than constantly demanded. More recent design work has examined how these principles can be applied in contemporary contexts. Hassan et al. (2024) explore how calm design strategies may be implemented through simple, low-intrusive cues that remain in the background of user activity. While their study focuses on interactive advertising, it demonstrates how calm principles continue to inform contemporary interaction design.

2.3 Intrusiveness as a Design Dimension

While Calm Technology frames reduced intrusiveness as desirable, empirical research suggests that system interventions may serve both disruptive and supportive functions depending on how they engage users' attention. Research in human-computer interaction suggests that interactions with digital systems may require different levels of user attention, ranging from peripheral awareness to focused engagement (Bakker & Niemantsverdriet, 2016). From this perspective, system notifications can be understood

as interaction events that vary in how strongly they draw user attention. This perspective also highlights that interaction is not static but involves continuous shifts between levels of attention. System interventions may therefore disrupt or support workflow depending on how they move the user between peripheral awareness and focused engagement, and how this aligns with the users' ongoing task.

Similarly, Iqbal and Horvitz (2010) found that notifications in workplace settings often function as mechanisms for maintaining awareness, but they may also trigger task switching depending on timing and relevance. Although users acknowledge notifications as disruptive, they often choose to keep them enabled because of the perceived value of staying informed. Turning notifications off does not uniformly improve focus; instead, users respond differently depending on their work patterns and awareness needs.

Together, with interruption research (Czerwinski et al., 2004; Nadj et al., 2023), these findings suggest that intrusiveness should not be understood as a binary quality. Rather, system intervention varies in intensity, timing, and form, and its experiential impact depends on how it aligns with task demands and cognitive state. This suggests that the goal is not to eliminate interruptions entirely, but to shape them more carefully.

This perspective becomes even more complex in adaptive systems, where system behaviour changes dynamically based on user data and contextual factors. Jingar (2025) describes how intelligent coaching systems rely on personal and dynamic adaptation to support behaviour change. While such adaptability may increase relevance, it also introduces challenges related to transparency and perceived control. When system behaviour changes automatically without clear communication, users may experience uncertainty regarding how and why the system intervenes. This suggests that intrusiveness is shaped not only by interruption intensity, but also by how adaptive behaviour is understood and experienced by users.

This chapter has presented the theoretical foundations of the study. Research on digital work interruptions demonstrates how system-initiated interventions affect workflow continuity and cognitive effort. Calm technology provides a perspective on how systems can manage attentional demand and move between foreground and background interaction.

Together these perspectives support understanding intrusiveness as a graded interaction quality rather than a binary condition. Intrusiveness can vary depending on how much attention a system demands, how it interrupts ongoing work and how much control the user retains.

Based on this understanding, the study treats intrusiveness as a design variable that can be systematically varied. This forms the foundation for the design scenarios presented in the next chapter.

The concepts presented in this chapter are used in three ways throughout the study. First, they inform the design of the scenarios by structuring different levels of interruption, attentional demand and user control. Second, they guide the development of the interview questions by focusing participants' reflections on experiences of focus, workflow and

perceived control. Third, they provide a conceptual framework for interpreting participants' experiences during the analysis phase.

3 Methodology

3.1 Research Approach

A qualitative approach was considered appropriate for this study because the aim was to gain a deeper understanding of how IT professionals experience different levels of intrusiveness in supportive system features during computer-based work. Rather than measuring objective outcomes such as productivity or task performance, the study focused on participants' perceptions and experiences of system interventions in relation to focus, workflow, and perceived control. Qualitative methods are particularly useful when the aim is to understand individuals' experiences and interpretations of a phenomenon (Jacobsen, 2015), which aligns with the purpose of this study.

3.2 Research through Design (RtD)

To explore these experiential qualities, the study adopts a Research through Design (RtD) approach, where design artefacts are used as a way of exploring and generating knowledge about interaction qualities rather than evaluating technical system performance (Zimmerman et al., 2007). The focus of this study is on how different levels of intrusiveness in supportive features shape user experience of focus, workflow, and perceived control in computer-based work environments.

Within experience-centred design and Research through Design, scenarios are often used to explore how technologies may be experienced in everyday contexts and to stimulate reflection on alternative interaction possibilities (Wright & McCarthy, 2010). In this study, conceptual design scenarios represent different levels of system intrusiveness and support participants' reflection on how such interventions might be experienced during focused computer-based work.

3.2.1 Design process and Data collection

The design process was organised into several phases in which the development of design scenarios and the collection of empirical data were closely connected. Conceptual design scenarios were developed as discussion artefacts representing different levels of system intrusiveness and were presented to participants during semi-structured interviews. This allowed participants to reflect on how such interventions might be interpreted and experienced in everyday computer-based work.

The process consisted of four main phases: exploration of the design problem, concept development, scenario prototyping, and evaluation with users.

3.2.2 Phase 1 — Exploration

To operationalise this conceptualisation within the design process, three dimensions: (1) degree of interruption of ongoing work, (2) level of attentional demand, and (3) degree of

user control over the intervention, were translated into the design criteria used to structure the scenarios developed in the following phase.
Degree of interruption refers to the extent to which the system interrupts or pauses ongoing work tasks. This criterion draws on research on digital interruptions, which shows that interruptions influence task resumption and workflow continuity (Czerwinski et al., 2004).
Level of attentional demand concerns whether the system requires focal attention or remains in the periphery of awareness. This distinction is informed by research on calm technology and peripheral interaction, which emphasises that systems may operate at different levels of attentional demand (Weiser & Brown, 1996; Hassan et al., 2024).
Degree of user control refers to whether the system enforces an action or allows the user to decide how to respond to the intervention. This dimension is inspired by research on notification management and interruption timing, which highlights the role of user control in how system interventions are experienced (Iqbal & Horvitz, 2010).
Together, these three criteria informed the construction of the design scenarios in the following phase, where the level of system intervention was systematically varied while the work context remained constant.

3.2.3 Phase 2 — Ideation and Concept Development

Building on the conceptualisation and design criteria identified in Phase 1, this phase focused on exploring how supportive system features might intervene in users' workflow at different levels of intrusiveness.
The aim of this phase was to translate the theoretical dimensions of intrusiveness, interruption, attentional demand, and user control into possible forms of system interaction that could support focused computer-based work.
During the ideation process, different potential forms of system intervention were explored by considering how a supportive system might vary in terms of:

- the degree to which it interrupts ongoing work
- the amount of attention it requires from the user
- the level of control the user retains over the intervention

3.2.4 Phase 3 — Prototyping

In this phase, the conceptual levels of intrusiveness identified in the previous phases were translated into structured design scenarios. The purpose of the scenarios was to represent different forms of system intervention during focused computer-based work.
To enable a meaningful comparison between the scenarios, the work situation and the supportive intention of the system were kept constant. Each scenario describes the same situation in which an IT professional is engaged in a complex coding task that requires sustained concentration. During the task, a system notification appears suggesting that the user take a short break.

While the context and the intention of the support remain the same, the degree of system intervention varies across the scenarios. The scenario therefore represents different levels of intrusiveness in how the system communicates the suggestion.

3.2.5 Phase 4 — Evaluation with Users

The scenarios were evaluated through semi-structured interviews with IT professionals. During the interviews, participants were presented with the scenarios and asked to reflect on how each scenario would influence their focus, workflow, and sense of control. The discussion also explored whether the support would feel helpful or disruptive, and under what conditions a system intervention might be experienced as intrusive.

The interviews lasted approximately 45–60 minutes and were audio-recorded with participants' informed consent. In addition, field notes were taken during the interviews to capture non-verbal cues and relevant contextual observations. The recordings were transcribed verbatim for analysis.

The study involved six IT professionals who primarily worked in computer-based environments. A purposive sampling strategy was applied, meaning that participants were selected based on their relevance to the research focus rather than through random recruitment. As described by Palinkas et al. (2015), purposive sampling is appropriate in qualitative research when the aim is to identify information-rich cases that can provide in-depth insight into the phenomenon under study.

Participants were recruited through the researcher's professional and personal networks. Individuals known to work in IT-related roles were contacted directly, and additional participants were identified through referrals from these initial contacts. This approach provided access to professionals who regularly engaged in focused computer-based work and who had experience with supportive or adaptive system features.

Inclusion criteria were as follows: working primarily in digital work environments and regularly performing tasks that require sustained attention. The aim of this sampling strategy was not to achieve statistical representativeness, but to obtain rich, experience-based insights relevant to the research focus.

3.3 Data analysis

The interview data were analysed using thematic analysis, following the six-phase approach outlined by Braun and Clarke (2006): familiarisation with the data, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the final report. Thematic analysis is a flexible qualitative method that enables the systematic identification and interpretation of recurring patterns within data.

The transcripts were coded and organised into themes that capture recurring patterns in how participants experience different levels of intrusiveness, with particular attention to focus, workflow, and perceived control. Field notes taken during and immediately after the interviews were also used to support interpretation during analysis. The analysis was

used to articulate design-relevant insights about how supportive features can be shaped to balance support and interruption in everyday work contexts.

3.4 Ethical Considerations

Ethical considerations were taken into account throughout the research process. The study followed the ethical guidelines for research involving human participants outlined by the Swedish Research Council (Vetenskapsrådet, 2017).

Prior to participation, all participants received written information about the purpose of the study, the procedures involved, and how the collected data would be used.

Participants were informed that their participation was voluntary and that they could withdraw from the study at any time without providing a reason.

All participants signed an informed consent form before the interviews were conducted (see Appendix 1). The interviews were audio-recorded only with participants' consent. To protect participants' privacy, all collected data were pseudonymised during transcription, and no identifying information was included in the final report. The recordings and transcripts were stored securely and used solely for research purposes.

4 Results

4.1 Phase 1 — Exploration

The exploration phase was based on previous research on interruptions, notifications, and calm technology. Studies on digital interruptions demonstrate that system-initiated alerts may disrupt ongoing work and increase the effort required to resume tasks (Czerwinski et al., 2004). Similarly, research on notification behaviour shows that the disruptive impact of notifications depends on their timing and salience (Iqbal & Horvitz, 2010).

Research on Calm Technology proposes that interactive systems should support peripheral awareness rather than constantly demanding focal attention (Weiser & Brown, 1996). More recent design research further explores how interactive systems can provide subtle and low-intrusive forms of support in the background of users' activities (Hassan et al., 2024).

Based on this synthesis of the literature, intrusiveness is conceptualised in this study as a design quality consisting of three dimensions: degree of interruption, level of attentional demand, and degree of user control. Together, they provided the foundation for structuring the design scenarios developed in the following phase.

4.2 Phase 2 — Ideation and concept development

Building on the design criteria identified in Phase 1, namely degree of interruption, attentional demand, and user control, this phase focused on exploring how different configurations of system intervention could be expressed in interaction design.

Through the ideation process, multiple potential interaction forms were explored by systematically varying how a system might interrupt ongoing work, demand user attention, and constrain or preserve user control. Attention was given to identifying interventions that would be perceived as more or less disruptive, ranging from those that fully interrupt workflow and require immediate action, to those that remain in the periphery and do not require any response.

From this broader set of possibilities, three configurations were selected to represent distinct and contrasting positions along the spectrum of intrusiveness. The selection was based on their ability to clearly differentiate levels of system intervention while maintaining conceptual clarity and comparability.

The first configuration represents a high level of intrusiveness, where the system fully interrupts the ongoing task, demands immediate attention, and overrides user control. The second represents a moderate level, where the system remains noticeable and requires attention while allowing the user to decide how to respond. The third represents a low level of intrusiveness, where the system operates unobtrusively in the periphery without interrupting the task or requiring any action.

These configurations were interpreted as representing high, moderate, and low levels of intrusiveness, forming a graded structure that captures variation in system intervention.

The decision to structure the design space into three levels was made to create a clear analytical contrast while maintaining a manageable scope for comparison. This structure enables intrusiveness to be explored as a graded interaction quality rather than a binary condition.

As shown in Table 1, multiple interaction concepts were explored and evaluated in relation to their level of interruption, attentional demand, and degree of user control.

Table 1

Explored interaction concepts and selection criteria across levels of intrusiveness.

Scenario	Explored alternatives	Selected concept	Rejected alternatives	Selection criteria
High intrusiveness	Full-screen popup with icon, sound alert, animated avatar	Full-screen popup with icon	Sound alerts and animated avatars were excluded to avoid additional sensory stimuli beyond visual intrusiveness.	Strong interruption and attention capture without sensory overload
Moderate intrusiveness	Small centre popup, side popup with avatar, popup with response options	Popup with ignore/postpone options	Centre popups were excluded because they reduced the distinction between moderate and high intrusiveness. Avatar-based popups attracted too much attention and risked increasing perceived disruption.	Balance between interruption and user control
Low intrusiveness	Small indicator, subtle animation, fading screen-colour notification	Peripheral subtle indicator	Animated elements and fading screen-colour notifications risked shifting attention from peripheral	Minimal disruption while preserving awareness and focus

	awareness to focal attention.

The explored alternatives illustrate how different interaction qualities were intentionally refined to support a comparative evaluation of intrusiveness. This phase resulted in the development of three conceptual levels of intrusiveness, which later formed the basis for the design scenarios presented in Phase 3.

4.3 Phase 3 — Prototyping

In this phase, the conceptual levels of intrusiveness identified in the previous phase were translated into three structured design scenarios. These scenarios act as discussion artefacts, representing different forms of system intervention during focused computer-based work.

To enable a meaningful comparison, the work context and the purpose of the system intervention were kept constant across all scenarios. Each scenario depicts the same situation: an IT professional engaged in a complex coding task requiring sustained concentration. During the task, the system presents a suggestion to take a short break. While the purpose of the intervention remains the same, the interaction quality varies across the scenarios based on the three design criteria: degree of interruption, level of attentional demand, and degree of user control. The following sections describe how these variations are expressed across the three scenarios.

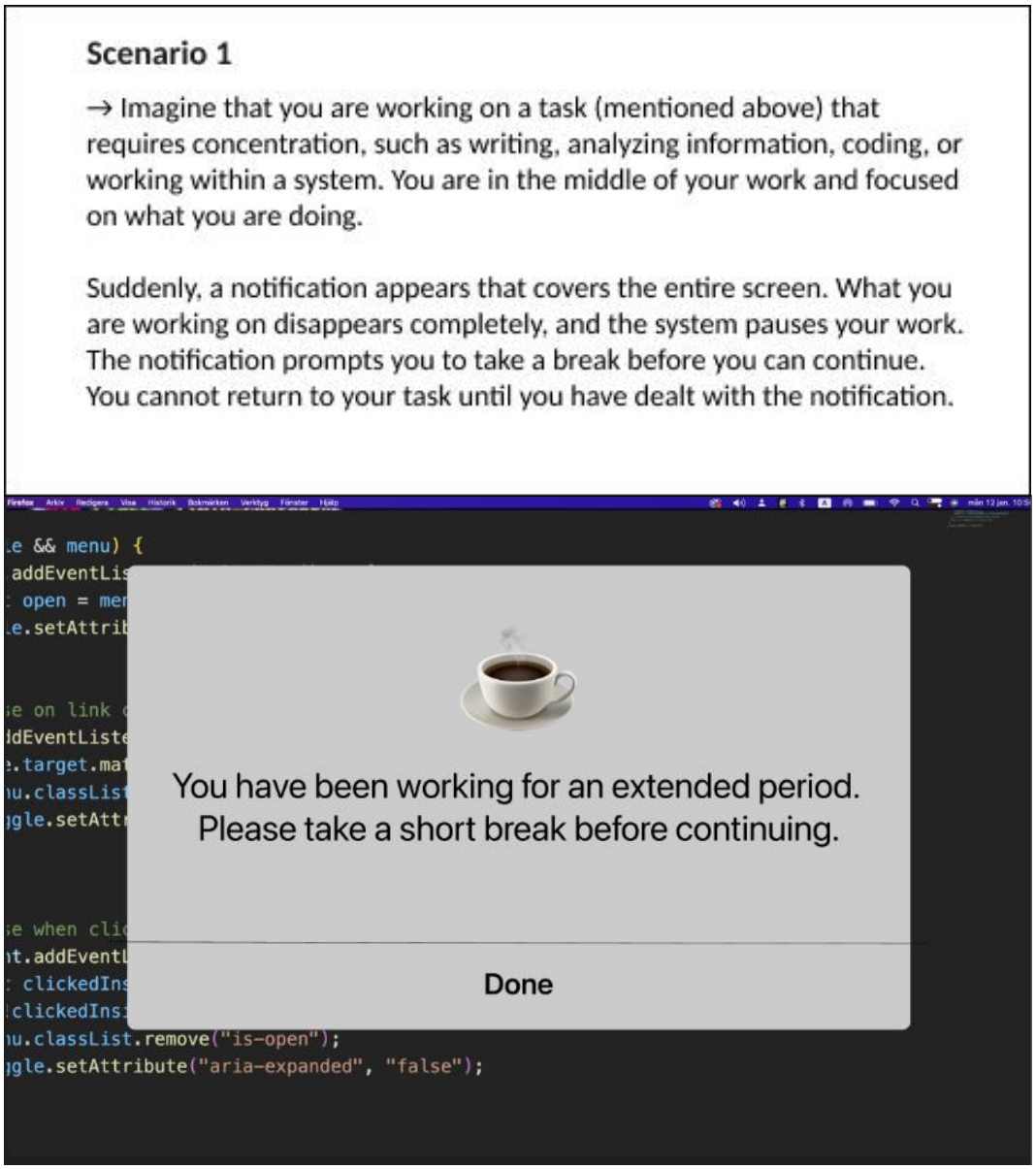
4.3.1 Scenario 1 — High Intrusiveness (full-screen notification)

At the high level of intrusiveness, the system fully interrupts the ongoing task, demands immediate attention, and temporarily overrides user control. While working on a complex coding task, the user is suddenly interrupted by a full-screen notification that covers all active windows, making it impossible to continue working. The notification suggests that the user take a break and requires a response before the task can be resumed. Until the notification is addressed, the system prevents any interaction with the work environment. A visual representation of the scenario is shown in Figure 1.

Figure 1

interpretation of Scenario 1, illustrating a high-intrusiveness full-screen notification.

Note. (Top) Scenario 1 as presented during the interviews. (Bottom) Visual mockup created by the authors in



Figma to support the reader's understanding of the scenario

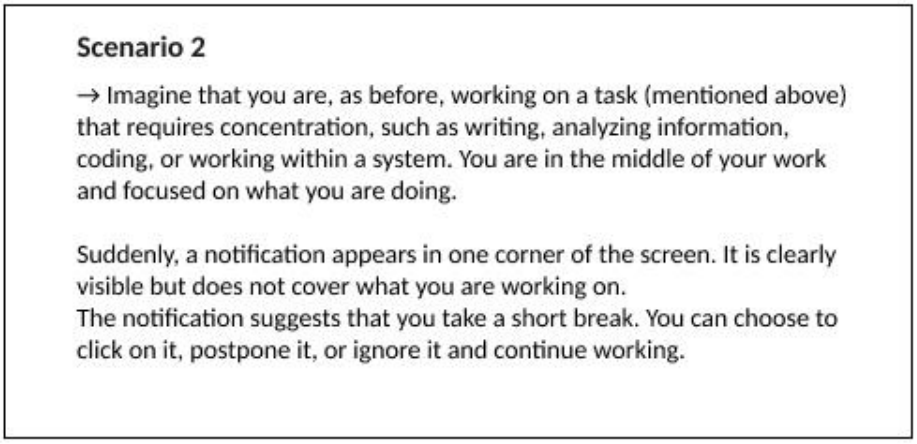
4.3.2 Scenario 2 — Moderate Intrusiveness (partial notification with options)

At the moderate level of intrusiveness, the system is designed to be noticeable without interrupting the ongoing task, requiring user attention while preserving control. While working on a complex task, a notification appears in the corner of the screen, suggesting that the user take a short break. Unlike the high-intrusiveness scenario, the notification does not block the work environment, allowing the user to continue working without being forced to respond.

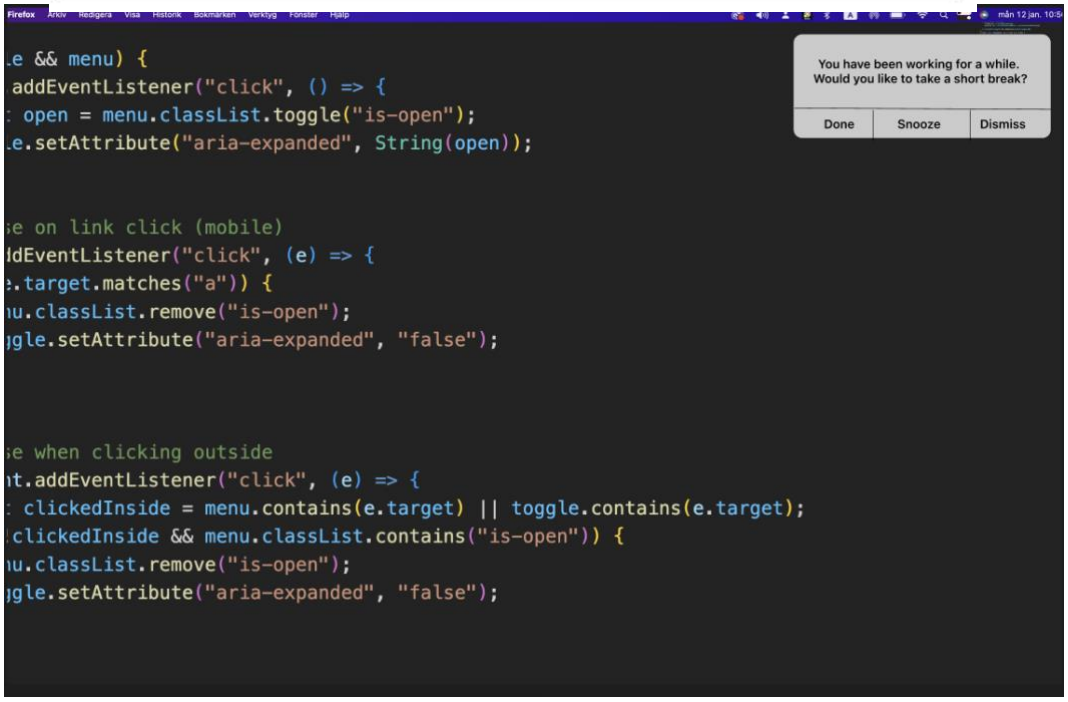
The notification remains visible and draws some attention, while the user retains full control and can choose to accept, postpone, or ignore the suggestion while continuing their work. A visual representation of the scenario is shown in Figure 2.

Figure 2

Visual interpretation of Scenario 2, illustrating a moderate-intrusiveness notification with response options.



Note.



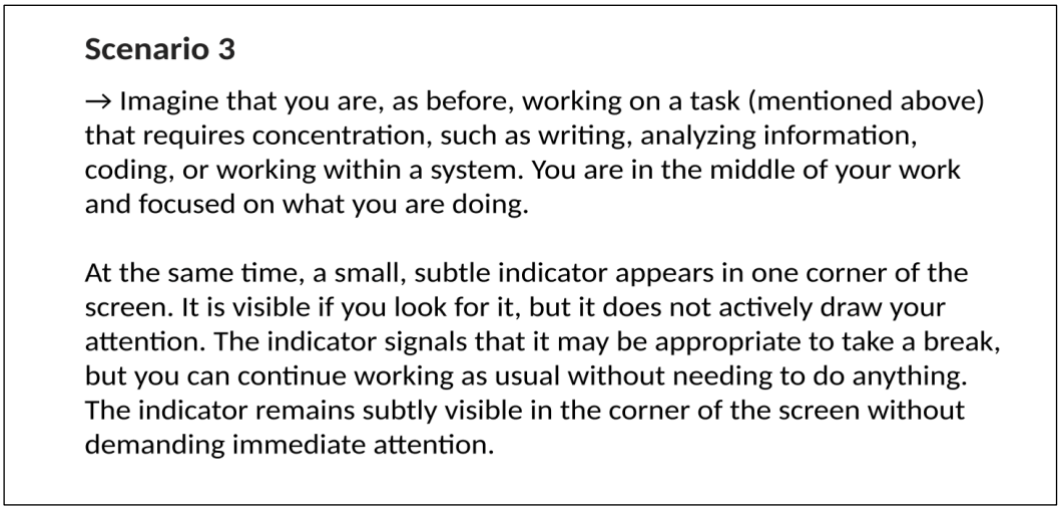
(Top) Scenario 2 as presented during the interviews. (Bottom) Visual mockup created by the authors in Figma to support the reader's understanding of the scenario

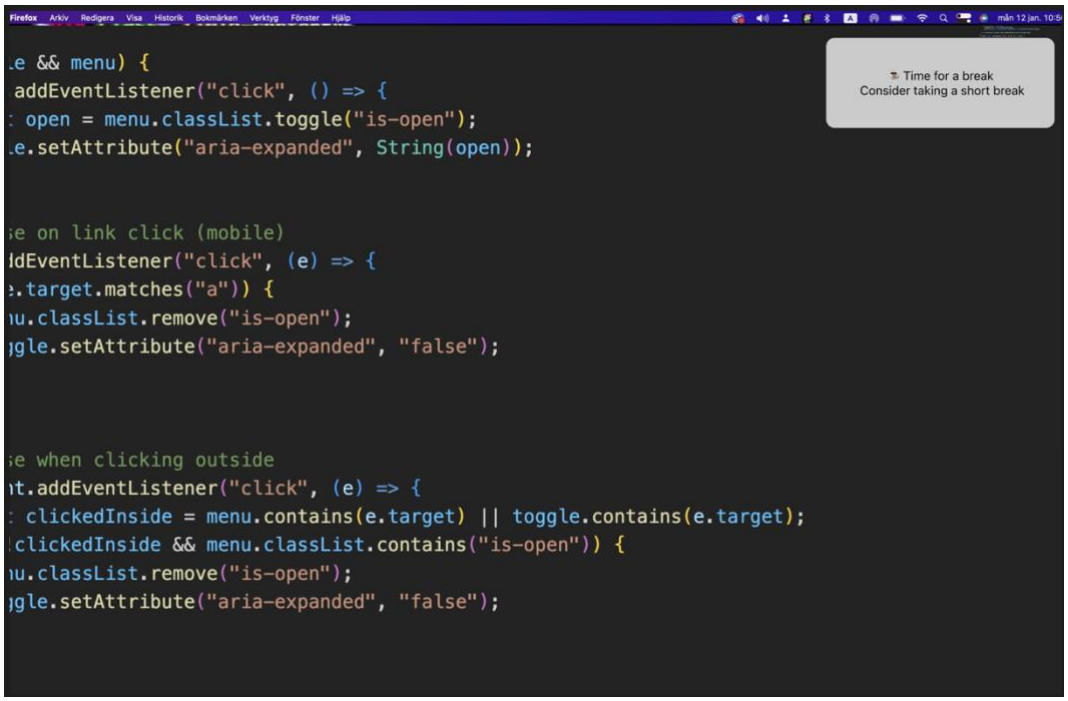
4.3.3 Scenario 3 — Low Intrusiveness (subtle status indicator)

At the low level of intrusiveness, the system is designed to minimise interruption, operate in the periphery of attention, and maintain full user control. While working on a complex task, a subtle indicator appears in the corner of the screen, suggesting that the user may take a short break. The indicator does not interrupt the ongoing task or require immediate attention, allowing the user to continue working without disruption. The user retains full control and can choose to respond to the suggestion at any time, while the system remains in the background without requiring any action. A visual representation of the scenario is shown in Figure 3.

Figure 3

Visual interpretation of Scenario 3, illustrating a low-intrusiveness notification designed to remain in the periphery of attention.





Note. (Top) Scenario 3 as presented during the interviews. (Bottom) Visual mockup created by the authors in Figma to support the reader's understanding of the scenario

4.4 Phase 4 — Evaluation with users

This phase focused on evaluating the design concepts through user involvement. The purpose of this phase was to explore how different levels of intrusiveness are experienced in real work contexts, and how such interactions influence users' focus, workflow, and sense of control.

A total of six participants were recruited, all of whom work in digital environments that require sustained focus and concentration. The participants represent different professional roles, including software developers, a product owner, a network specialist, and a data analyst. This diversity provided a broad perspective on how system interruptions are experienced across different types of work. The interviews lasted between 35 and 50 minutes and were conducted individually. During the interviews, participants were asked to describe their immediate reactions, perceived impact on focus and workflow, and their sense of control in relation to each scenario. Table 2 provides an overview of the participants, including their roles, background, and selected example quotes.

Table 2

Overview of participants and selected interview quotes related to experiences of intrusiveness.

Interviews	Role & Background	Time	Quotes
P1 Swedish speaker	Software Developer High need for focus	45 min	"Om något dyker upp i hörnet tar det fokus och avbryter tankebanan man befinner sig i." ("If something appears in the corner of the screen, it shifts my attention and interrupts my train of thought.") Related to moderate intrusiveness
P2 Swedish speaker	Product Owner Works with product decisions High need for focus	45 min	"Jag vill kunna bestämma själv .. så att jag blir lagomt störd." ("I want to decide myself .. so that I become interrupted just enough") Related to moderate intrusiveness
P3 Swedish speaker	Software Developer High need for focus	35 min	"Frånkopplad från verksamhetens behov" ("Disconnected from business needs") Related to moderate intrusiveness
P4 Swedish speaker	Software Developer High need for focus	40 min	"Det är som att någon annan kommer in i rummet och stör mig." ("It feels like someone enters the room and interrupts me."). Related to: High intrusiveness
P5 English speaker	Network Specialist High need for focus	50 min	"You lose your flow… and it's hard to get back to the same concentration." Related to High intrusiveness "It's like a friend telling you, 'do you need a break?'" Related to: Low intrusiveness

P6 Swedish speaker	BI Analyst Works with focus-intensive and sensitive tasks	35 min	"Det ger mig en dålig känsla… att jag borde tänka på min hälsa." ("It gives me a bad feeling… like I should take better care of myself.") Related to moderate intrusiveness

4.4.1 Scenario 1 — High Intrusiveness (full-screen notification)

Participants emphasized that forced interruptions made it difficult to continue working, especially in tasks that require deep concentration. Losing focus was described as particularly problematic, as it was not always easy to return to the same level of engagement after the interruption. One participant compared this type of interruption to someone *"entering the room and interrupting me"*, highlighting how intrusive and disruptive it felt.

One participant described this as losing their mental flow: *"You lose your flow… and it's hard to get back to the same concentration."*

Importantly, participants also highlighted the high recovery cost of such interruptions. One participant explained that being interrupted during complex problem-solving could result in losing up to 20 minutes of effective work time, due to the effort required to regain focus.

A central issue in this scenario was the lack of control. When participants felt that they could not ignore or postpone the notification, the system was experienced as taking over their workflow. This created frustration and, in some cases, increased stress. One participant described this as "*100% disruptive*", emphasizing the severity of the experience.

The impact of these interruptions was also influenced by task type and frequency. Frequent interruptions were described as more disruptive and stressful, while occasional interruptions were seen as slightly more manageable. In addition, participants questioned the relevance of such interruptions in real work situations.

There was some nuance in the responses. One participant reflected on the possibility of adapting to such systems over time, suggesting that users might gradually adjust their behaviour. Another participant noted that similar interventions could be acceptable if they were self-initiated, for example as part of a structured "focus module", rather than enforced by the system. Overall, high intrusiveness was experienced as unsuitable for focused work, due to its strong negative impact on focus, workflow, and perceived control.

4.4.2 Scenario 2 — Moderate Intrusiveness (partial notification with options)

The moderate level of intrusiveness was generally perceived as acceptable, but not without some drawbacks. Participants described these notifications as manageable because they allowed flexibility and did not force immediate action.

A key aspect influencing the experience was the presence of options, such as ignoring or postponing the notification. This gave participants a sense of control and allowed them to decide how to respond based on their current task and level of focus. However, even when the notification was not forced, it could still create momentary distraction. Some participants described feeling curious about the notification, which briefly shifted their attention away from the task.

Importantly, participants also described how these interruptions affect their cognitive flow. One participant explained that notifications break their *"train of thought"*, while another highlighted that even small interruptions can disrupt a state of deep concentration, or being *"in the zone"*.

Another important finding was that the experience depended on the type of task. During complex or focus-demanding work, the notification was more likely to be perceived as disruptive. In contrast, during simpler tasks, participants were more open to engaging with the notification and taking breaks. One participant described an emotional response, where the notification triggered a feeling of guilt when ignoring well-being reminders: "*It gives me a bad feeling… like I should take better care of myself."*

At the same time, some participants expressed a preference for managing breaks themselves rather than relying on system interventions, highlighting the importance of autonomy in how interruptions are handled.

Moderate intrusiveness was experienced as acceptable and sometimes helpful, as long as users retained control over the interaction.

4.4.3 Scenario 3 — Low Intrusiveness (subtle status indicator)

Low levels of intrusiveness were consistently perceived as the most supportive and least disruptive. Participants described these notifications as subtle and easy to ignore, allowing them to remain focused on their work. Because the notification did not require immediate action, it did not interrupt workflow. Instead, it provided a form of peripheral awareness, where users could decide whether and when to respond.

Several participants highlighted that this type of interaction gave them a strong sense of control. The ability to ignore the notification without consequences was central to its positive experience. In addition, the notification was sometimes described as supportive, especially during less demanding tasks. In these situations, it could act as a gentle reminder to take a break and maintain well-being. Some participants described this as a *"nice reminder"* that is not bothersome, as it allows them to stay in control of their workflow.

One participant compared this type of notification to a social interaction rather than a system enforcing an action: *"It's like a friend telling you, 'Do you need a break?'"*
This highlights how low-intrusive interactions can be perceived as more natural and human-like, where the system supports rather than interrupts the user. Participants also emphasized that this type of design allows them to engage only when it fits their workflow.
In addition, one participant noted that this interaction aligns with their current way of working, where they rely on subtle indicators, such as small badges in applications, which they check at their own discretion. This reinforces how low-intrusive notifications support existing user behaviour rather than disrupting it.
However, a potential limitation was also identified. Because the notification is subtle, there is a risk that it may go unnoticed, particularly during periods of deep focus. Another important insight was the role of context. Even subtle notifications could become distracting in shared work situations, such as meetings or screen sharing.
Overall, low intrusiveness was experienced as the most effective approach, balancing awareness and minimal disruption while maintaining user control.

4.5 Comparative findings

Across all participant interviews, clear patterns emerged in how different levels of intrusiveness were experienced in relation to focus, workflow, and perceived control.
High-intrusive interactions were consistently described as disruptive. Participants emphasized that forced interruptions, especially those that required immediate action or covered the screen, broke their concentration and made it difficult to return to the task. These types of interactions were also associated with frustration and, in some cases, increased stress, particularly during tasks requiring deep focus.
Moderate levels of intrusiveness were generally perceived as more acceptable. Participants highlighted that having options, such as ignoring or postponing the notification, allowed them to manage the interruption based on their current situation. This flexibility supported a sense of control and made the interaction less disruptive, although it could still cause momentary distraction. In some cases, moderate notifications also triggered emotional responses, such as guilt related to well-being reminders, even when they did not directly interrupt the task.
Low-intrusive interactions were consistently perceived as the most supportive. Participants described these notifications as non-disruptive, allowing them to remain focused while still being aware of the suggestion. The ability to ignore the notification without consequences was a key factor in maintaining control and workflow continuity. Some participants described this type of interaction as more natural and even human-like, contributing to a more positive user experience. However, the findings also highlight important nuances.

The effectiveness of low-intrusive interactions depends on whether they are noticed, as subtle notifications risk being overlooked. Additionally, the experience of intrusiveness was influenced by context, including workload and social situations such as meetings or shared screens. Several participants also reflected that the perceived intrusiveness of an intervention may vary depending on the nature of the ongoing task. While the scenarios focused on work requiring sustained concentration, many suggested that interruptions might be experienced differently during more routine tasks compared to tasks requiring deep focus.

4.6 Design preferences

Across the interviews, several consistent design preferences emerged regarding how system notifications should be designed to support users' focus and workflow. A central preference was maintaining user control. Participants consistently expressed the need to be able to ignore, postpone, or engage with notifications at their own discretion. Forced interactions, where the system required immediate action, were strongly rejected. Another key preference was subtlety. Participants preferred notifications that are visually unobtrusive and placed in the periphery of attention, rather than covering the screen or interrupting ongoing tasks. Small indicators or notifications that disappear automatically were described as more acceptable and less disruptive.

Timing and context-awareness were also highlighted as important factors. Participants emphasized that notifications should adapt to the user's current activity, avoiding interruptions during tasks that require deep concentration or during meetings. Some participants suggested that systems could integrate with calendars or detect user activity to reduce inappropriate interruptions.

Flexibility and customisation were further identified as desirable features. Participants expressed a preference for being able to adjust aspects such as notification size, placement, duration, and the presence of sound. This would allow the system to better align with individual preferences and different work environments.

Finally, participants described a preference for more human-like and supportive interactions. Subtle notifications were sometimes perceived as similar to a "friendly reminder" rather than a command. This type of interaction was experienced as more natural and less stressful.

5 Analysis

5.1 Thematic Analysis Approach

This chapter presents the analysis of the interview data using a thematic analysis approach inspired by Braun and Clarke (2006). The aim of the analysis is to move beyond describing participants' responses and instead interpret underlying patterns in how different levels of intrusiveness are experienced in digital work environments.
Across the data, three main themes were identified: control, disruption, and focus. These themes capture central aspects of how users experience system interventions, particularly in relation to maintaining workflow and concentration. Rather than analysing each scenario separately, the themes are explored across all three levels of intrusiveness, high, moderate and low in order to identify patterns and differences in user experience.
This approach allows for a more comprehensive understanding of intrusiveness as a dynamic and context-dependent quality. The analysis also draws on relevant research to interpret the findings, including studies on interruptions and attention management, as well as concepts from Calm Technology.

5.2 Control

Control emerged as a central factor influencing how users experience system interruptions across all levels of intrusiveness. The findings show that the perception of control varies significantly depending on how the interaction is designed.
In the high intrusiveness scenario, perceived control appeared closely connected to the user's ability to regulate interruptions independently. When the system demanded immediate attention, participants experienced a reduced sense of autonomy and greater disruption to their ongoing work. This suggests that forced interactions may disrupt both task continuity and users' ability to manage attention on their own terms.
In contrast, the moderate intrusiveness scenario presents a more complex and less consistent pattern. While the system provides users with options, such as ignoring or postponing the notification, perceived control is not uniform. Instead, responses were distributed across positive, acceptable, and negative evaluations.
These findings indicate that the presence of options does not automatically result in a clear sense of control. Instead, control depends on how the interruption aligns with the users' current task, where the ability to ignore or postpone the interruption can support autonomy, but may still be experienced as intrusive during periods of high concentration. Rather than representing a middle ground between high and low intrusiveness, the moderate scenario reflects a more polarized experience of control, where individual differences and task context influence how the interaction is perceived.
The low intrusiveness scenario is consistently associated with a strong sense of control. Because the notification does not require immediate action and can be ignored without

consequence, it allows users to maintain flexibility and remain undisturbed in their workflow. Participants described it as flexible and non-intrusive.
These findings highlight that control is not only about having options, but also about when and how those options are presented. When users are able to maintain control over workflow, interruptions are more likely to be perceived as supportive rather than disruptive. This aligns with previous research on attention and interruptions. Iqbal and Horvitz (2010) argue that the ability to defer interruptions plays a key role in reducing their negative impact. The findings in this study support this perspective, suggesting that when users can manage interruptions on their own terms, the experience becomes significantly more positive.

5.3 Disruption

Disruption emerged as a central theme in participants' experiences of system interruptions, particularly in relation to maintaining focus and ongoing work. Participants described different levels of disruption depending on how intrusive the interaction was, highlighting how interaction design shapes the experience of interruptions.
In the high intrusiveness scenario, participants consistently described the interaction as highly disruptive. Interruptions that required immediate action and blocked access to ongoing work were perceived as breaking the workflow entirely. Participants emphasized that such interruptions not only stopped the task but also created difficulties in resuming it. For example, P5 described how interruptions disrupted workflow continuity: "You lose your flow… and it's hard to get back to the same concentration."
Similarly, P4 compared the interruption to "someone entering the room and interrupting me," highlighting how intrusive the full-screen notification was experienced during focused work. The disruption was described as both cognitive and practical, as users had to reorient themselves to the task after interruption.
In the moderate intrusiveness scenario, disruption was present but less severe. Although participants had the option to ignore or postpone the notification, the interruption still created a momentary shift in attention. Even when users were given control over the notification, the interruption still created a temporary shift in attention. However, the impact was more manageable, as users could quickly return to their task without being forced to disengage completely.
In contrast, the low intrusiveness scenario was rarely experienced as disruptive. Because the notification remained in the periphery and did not require immediate interaction, it allowed users to maintain their workflow. P5 described the interaction as: "It's like a friend telling you, 'Do you need a break?'" and other participants described these interactions as non-intrusive and easy to ignore, which minimised their disruptive impact.
These findings align with previous research on interruptions in digital work environments. Czerwinski et al. (2004) show that interruptions can significantly disrupt task performance, particularly when they occur during cognitively demanding work. The

difficulty of resuming a task after an interruption, often referred to as the "resumption lag", is a key factor in how disruptive an interruption is perceived to be. The findings in this study support this perspective, as participants described challenges in returning to their tasks after high-intrusive interruptions, while low-intrusive interactions minimised this effect. This suggests that disruption is not only about the presence of an interruption, but about how much it interferes with the continuity of work.

5.4 Focus

Focus proved to be a critical theme in understanding how users experience system interruptions in digital work environments. Across the interviews, participants emphasized the importance of maintaining concentration, particularly during tasks requiring sustained attention and cognitive effort.

In the high intrusiveness scenario, interruptions were experienced as highly disruptive to focus because the system forced participants to shift their attention away from the ongoing task during focused work. P1 described how forced interactions break their "train of thought" and interrupt their ability to remain in a state of deep concentration. This was especially important in tasks that require sustained mental engagement, such as coding or problem-solving, where participants felt they needed to keep multiple thoughts and processes active at the same time.

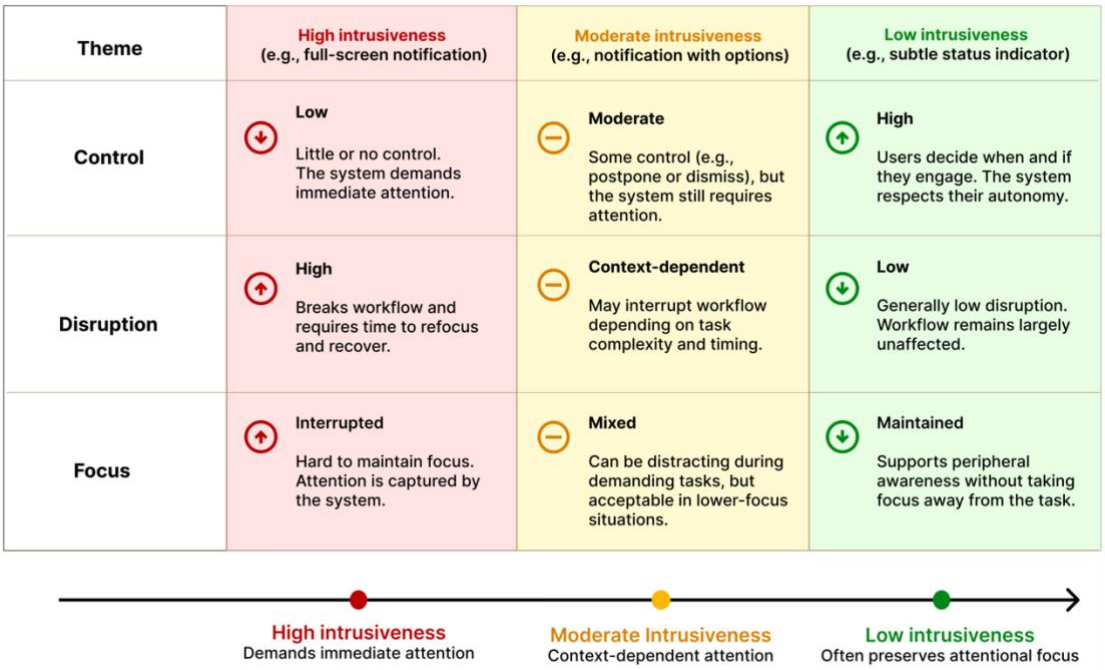
The disruption was not only connected to the moment when the notification appeared, but also to the effort required to regain focus afterwards. Participants explained that once their concentration had been interrupted, they often needed time to remember where they were in the task and mentally reconnect to their workflow. This suggests that highly intrusive interventions may create a cognitive recovery process that extends beyond the interruption itself.

In contrast, the low intrusiveness scenario was experienced as supportive of focus. Because the notification remained subtle and did not demand immediate attention, it allowed users to maintain concentration while still being aware of the system's presence. Participants described these interactions as operating in the background, enabling them to continue their work without interruption.

This distinction reflects principles from Calm Technology, as introduced by Weiser and Brown (1996), where technology should move between the centre and the periphery of attention. In this study, low-intrusive interactions remained in the periphery, supporting awareness without disrupting focus. In contrast, high-intrusive interactions forced themselves into the centre of attention, interrupting the user's cognitive process. The findings suggest that preserving focus requires designing systems that respect the user's attentional state. Rather than demanding attention, systems should provide information in a way that allows users to decide when to shift their focus. As illustrated in Figure 4, experiences of intrusiveness varied across the themes of control, disruption, and focus. High-intrusive interactions were generally experienced as disruptive and demanding,

while low-intrusive interactions were perceived as less disruptive and more supportive of workflow continuity.**Figure 4**

A comparative overview of how the three levels of intrusiveness were experienced across the main analytical themes identified in the thematic analysis.



Note. Comparative visualization created by the authors in Figma based on the thematic analysis findings.

6 Discussion

This chapter answers the research question and discusses the findings in relation to the research question, previous research and the broader field of interaction design. The discussion also reflects on the contributions and limitations of the study, as well as how the RtD approach shaped the knowledge through the project.

6.1 Answering the Research Question

This study aimed to explore how different levels of intrusiveness influence users' experiences of system interventions in digital work environments. The findings indicate that intrusiveness is not a fixed characteristic of a system, but rather something that depends on the situation. It is shaped by how the system behaves, the level of control given to the user, and the type of task being performed. Across all themes, low intrusiveness was generally perceived as the most supportive within the dimensions explored in this study, particularly in relation to focus and perceived control. However, the findings also showed that experiences of intrusiveness varied depending on the users' situation, task and attentional state.

Participants often described low-intrusive interventions as remaining in the background while still providing awareness of the system. This can be understood as a form of peripheral awareness, where users remain aware of information without being forced to shift their full attention. As a result, they can maintain focus and continue their work without unnecessary interruption. At the same time, the strong sense of user control associated with low intrusiveness also raises questions about how noticeable a supportive intervention needs to be in order to be perceived as useful rather than simply fade into the background.

In contrast, high-intrusive interactions disrupt this balance by forcing attention shifts and reducing the user's ability to manage their own workflow. Participants consistently described these interventions as disruptive and often associated them with frustration, loss of control and difficulties maintaining focus.

The moderate level of intrusiveness produced the most nuanced findings. Participants often accepted a more noticeable interruption when it was perceived as relevant and justified in relation to their current activity. The experience was influenced not only by the level of intrusiveness itself but also the intervention appeared to be meaningful, understandable and aligned with the users' situation. Unlike the high and low scenarios, the participants did not respond to the moderate intervention in a consistent way.

Overall, the findings suggest that designing for digital well-being is not about removing interruptions entirely but about shaping them carefully. Instead of minimising interruptions altogether, the results point towards the value of interventions thar are sensitive to context, support awareness and allow users to remain focused and in control.

6.2 Discussion of findings in relation to Research question

The aim of this study was to explore how varying levels of intrusiveness influence users' experiences of focus, workflow continuity and perceived control in computer-based work environments.

Evaluating graded levels of intrusiveness revealed that intrusiveness is not experienced as a fixed system property, but as a relational interaction quality shaped by context, attentional state and workflow. For example, disruption and support are not mutually exclusive but can coexist depending on how the interaction aligns with the users' current needs.

This relational nature of intrusiveness is also reflected in interruption research (Czerwinski et al., 2004; Iqbal & Horvitz, 2010), where system interventions often simultaneously provide awareness and disrupt workflow continuity depending on timing and task context. Significant individual differences emerged regarding users' tolerance for interruptions and their strategies for managing attention and workflow. While some preferred uninterrupted work, others are more open to system support, especially in situations where they need assistance in maintaining focus.

In some situations, the intervention also appeared to introduce an additional reflective demand, where users had to evaluate whether they should respond to the suggestion or continue working. This became visible in the moderate scenario, where one participant described feelings of guilt related to ignoring well-being reminders. The interruption therefore did not only concern attentional disruption, but also the mental effort involved in evaluating the intervention itself.

In addition to introducing reflective demands, the moderate scenario also generated more varied responses than the high and low intrusive scenarios. While the high-intrusive interactions were more clearly experienced as disruptive and the low-intrusive interactions were generally perceived as non-disruptive, the moderate level revealed a more nuanced and context dependent pattern. The same type of notification could be experienced as either supportive or disruptive depending on the user's current task, level of focus and surrounding situation.

These responses suggest that perceived control involves more than simply having access to options such as postponing or ignoring an interaction. Even when users were given a degree of control, the intervention could be experienced as intrusive if it interrupted workflow continuity or demanded attention at an inappropriate moment, such as during meetings, collaborative situations or periods of deep focus. At the same time, some participants described the moderate scenario as potentially helpful, particularly in situations where they experienced reduced focus or difficulty maintaining productivity. In these cases, the intervention was perceived less as a disruption and more as a supportive form of redirection or reset.

This preference for non-intrusive support aligns closely with principles of Calm Technology (Weiser & Brown, 1996), that technology is intended to remain closer to the

periphery of attention rather than constantly demanding focus. Participants generally responded more positively to interactions that allowed them to maintain awareness without fully interrupting their workflow. Yet, the findings also suggest that supportive interactions cannot simply remain invisible or passive at all times. In relation to the research question, this demonstrates that intrusiveness is shaped less by the interruption itself and more by how the intervention is experienced within the users' ongoing activity.

6.3 Contribution to Interaction design knowledge

This work contributes to interaction design knowledge by offering a deeper understanding of intrusiveness as a contextual and relational interaction quality. The main contribution of this study is the understanding that intrusiveness requires calibration rather than being treated as a fixed design property. This aligns with Bakker and Niemantsverdriet´s (2016) interaction-attention continuum, which suggests that interaction can occur at different levels of attention rather than requiring a fixed level of user engagement. Instead of simply reducing interruptions, the findings indicate that the experience of intrusiveness is shaped through the relationship between interventions, users and situational conditions.

Depending on the situation; the user's current task, stress level, and the timing of the intervention, users appeared more or less receptive to notifications and supportive features. In situations where the timing did not align with the user's current workflow or focus, the intervention was more likely to be experienced as intrusive. In contrast, interventions that appeared at more appropriate moments were perceived as more supportive and less disruptive. This aligns with previous research suggesting that interruptions are experienced differently depending on task relevance and situational context (Iqbal & Horvitz, 2010; Nadj et al., 2023). It became particularly visible in the moderate level of intrusiveness, where participants responded to the same interventions in different ways.

Interventions designed to be supportive may therefore become more disruptive when they interrupt cognitively demanding tasks at inappropriate moments. The participants' reflections reveal that intrusiveness depends less on the visual design itself, and more on the exact moment the interaction occurs.

Focused computer-based work was also found to be particularly sensitive to interruptions, especially during cognitively demanding tasks where participants described difficulties returning to the same level of concentration after being interrupted. So, the interruption itself is not the only challenge, but also the recovery effort required to return to the previous workflow and level of concentration. Similar observations were reported by Czerwinski et al. (2004), who found that interruptions increase the effort required to resume complex tasks after an interruption. This may also reinforce an underlying concern about losing focus, momentum or workflow continuity.

From an interaction design perspective, this emphasises that digital well-being systems cannot rely on static boundaries. Instead, the design knowledge generated through this study points toward the potential value of systems capable of adapting interventions to users' workflow and attentional state. This aligns with previous research suggesting that attention and interaction shift dynamically depending on context and cognitive state (Bakker & Niemantsverdriet, 2016; Iqbal & Horvitz, 2010). This could include detecting behavioural cues such as the frequency of tab switching or learning an individual's typical work tempo to avoid breaking the fragile state of deep focus. Future interaction design could benefit from focusing on creating systems capable of understanding the situational context to dynamically calibrate both the optimal timing and the appropriate visual presentation of the intervention.

A central aspect of this flexibility appears to be perceived control. Across the scenarios users were less likely to feel that the system was intrusive if they retained the possibility to ignore, postpone or decide when to engage with the system. The need for perceived control also became visible in how participants described their current strategies for managing notifications and interruptions in everyday work. Since no single system or notification structure fully matched their individual needs, several participants had developed their own routines by combining different tools, settings and personal strategies to better support focus and protect their workflow.

From an interaction design perspective, the findings suggest that eliminating interruptions entirely may not always be desirable but rather consider how different forms of intervention can be calibrated in ways that better align with the users' ongoing activity and context. The findings indicate that supportiveness and intrusiveness cannot be treated as opposites, since even noticeable interventions may be experienced as meaningful when they align with the users' current situation and allow flexibility in how the interruption is managed.

Instead of designing for a single optimal level of interruption, the study points toward the importance of adaptable and context-sensitive approaches where users are able to regulate how the supportive features appear and when they demand attention. In this sense, experiences of intrusiveness appear to depend on how system behaviour aligns with the user's expectations and current situation.

6.4 Discussion of Method

Intrusiveness can be difficult to discuss in abstract terms, since experiences of interruption are often shaped by context, task and attentional state. Without a concrete situation, discussions also risk being reduced to general opinions about notifications or interruptions. By creating structured scenarios and keeping the work situation constant while varying the level of intrusiveness, participants were able to compare and reflect on subtle experiential differences that may otherwise have remained difficult to articulate.

This highlights a key strength of the Research Through Design approach, where designed artefacts are used to explore and discuss possible experiences rather than evaluate a finished solution. By using scenarios as discussion artefacts, participants were able to move beyond general opinions about notifications and instead discuss how different forms of system behaviour might influence focus, workflow and perceived control in specific work situations (Zimmerman et al., 2007).

A more fully developed and interactive prototype may have provided a more explicit understanding of how users behave in relation to specific interventions and interaction flows. Such an approach could potentially have generated more detailed insights into which interactions worked well or caused friction. At the same time, the conceptual nature of the scenarios appeared to open up for broader reflections around intrusiveness, allowing participants to speculate, compare and reason about different forms of interruption across situations rather than reacting only to a single implemented solution. This was particularly visible in the moderate level of intrusiveness, where participants did not interpret the intervention in the same way. Instead, the discussion began shifting toward questions of timing, workflow, and perceived control where the same interaction could be experienced as either supportive or disruptive depending on the surrounding situation and attentional context.

Since experiences of interruption and loss of focus can be sensitive and emotionally charged, a fully implemented prototype may also have produced stronger immediate reactions of frustration or annoyance. While such reactions could have narrowed the conversation toward immediate emotional responses rather than deeper reflections about why certain interventions were experienced as intrusive in different contexts.

6.5 Limitations

As a Research through Design study, the purpose was not to produce statistically generalisable findings, but to generate design-oriented insights into how different forms of system intervention may be experienced in focused computer-based work. In line with RtD, the value of the study therefore lies less in universal claims and more in the potential transferability of the generated design knowledge to similar design contexts concerned with attention, interruptions and digital well-being (Zimmerman et al., 2007; Cash et al., 2022).

The study focused specifically on IT professionals with cognitively demanding computer-based work where workflow continuity and sustained focus were highly valued. While the limited number of participants reduces statistical representativeness, the intention of the sampling strategy was to obtain rich reflections from participants working in environments, where focus-intensive work formed a central part of everyday practice. Other forms of computer-based work may involve more fragmented workflows, collaborative environments or frequent meetings where interruptions are already integrated into everyday work practices. In such contexts, supportive interventions may

potentially be interpreted differently or become more accepted as part of the workflow itself.

The conceptual nature of the scenarios also shaped the type of knowledge produced in the study. Rather than evaluating a fully implemented system, participants reflected on conceptual scenarios representing different levels of intrusiveness. More developed low-fidelity or interactive prototypes may have generated stronger situated reactions and provided additional insights into how users respond to interruptions in practice. At the same time, the scenario-based approach appeared to support broader reflections around workflow, timing and perceived control, allowing participants to compare and reason about interruptions across different situations rather than reacting only to a single implemented solution.

Finally, the study was conducted within a limited timeframe and involved a relatively small number of design interactions. Additional RtD iterations, alternative scenario structures, or evaluating using more interactive artefacts may have further challenged the findings. Future RtD cycles could specifically investigate how proactive well-being systems can dynamically shift between the periphery and the centre of users' attention– moving away from fixed boundaries or static ambient states, to explore the fine line between helpful calibration and intrusive disruption.

As highlighted by Cash et al. (2022), the validity of design research depends on how clearly the boundary conditions of the study are articulated. The findings should therefore be understood within the specific context in which the scenarios were developed and evaluated, while still offering transferable reflections for future interaction design work related to supportive and low-intrusive systems.

7 Conclusion

The purpose of this thesis was to explore how different levels of intrusiveness in supportive system features shape IT professionals' experiences of focus, workflow continuity and perceived control during focused computer-based work.

To fulfil this purpose, our study contributes to a more nuanced understanding of intrusiveness by showing that supportive interventions cannot be understood as inherently disruptive or supportive themselves. Instead, the experience of intrusiveness appeared to emerge through the relationship between system behaviour, users' attentional state, and the surrounding work context.

The complexity of this relationship became particularly visible in the moderate level of intrusiveness, where participants responded to the same intervention in different ways. Unlike the high and low scenarios that generated more consistent reactions, the moderate intervention exposed variation in how support was interpreted and accepted. For some participants, interruptions functioned as helpful reminders or opportunities to reset attention, while others experienced similar interventions as disruptive when they conflicted with ongoing focus or workflow continuity. This indicates that support and disruption cannot be understood as opposite states, but as experiences shaped by situational demands, individual preferences, and perceived control.

Across the interviews, workflow continuity appeared highly valued during focused computer-based work. Participants often described focus as fragile. Interruptions created challenges not only in the moment of disruption itself, but also through the effort required to regain concentration and resume ongoing work. At the same time, some participants had already developed personal routines by combining different tools, settings, and strategies to manage notifications and protect focus. Such adaptive strategies illustrate how users actively shape technology around their own needs when existing systems do not fully align with individual workflows and attentional demands. This may partly reflect the participants familiarity with digital systems and work environments.

Taken together these insights contribute to interaction design knowledge by offering a deeper understanding of intrusiveness as a contextual and relational interaction quality instead of a fixed system property. The findings point toward the importance of calibrated intrusiveness, where supportive systems balance attentional awareness, workflow continuity and user autonomy in relation to the user's current context and focus.

7.1 Future research

Future research could explore how adaptive systems might calibrate different levels of intrusiveness based on users' current activity, attentional state and workflow. While this study used conceptual scenarios to investigate experiences of intrusiveness, future studies could evaluate fully implemented prototypes in real work settings. Such research could provide deeper insights into how supportive interventions are experienced over time and

how contextual factors influence the balance between support, disruption and perceived control.

The findings also highlight the nuances nature of the moderate level of intrusiveness, which generated the most varied responses among participants. Future research could explore this level in greater depth to better understand how supportive interventions can be designed to balance awareness, support and disruption in digital work environments. Such work may contribute to a clearer understanding of when interventions are experienced as acceptable, helpful or intrusive, and help inform the design of future digital well-being systems.

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Linnæus University

Appendix 1: Informed Consent form

Consent Form – Bachelor's/Master's Thesis Project

About the Study

The purpose of this study is to investigate how supportive features in digital work environments are experienced by users engaged in computer-based work. The study particularly focuses on how different levels of intrusiveness (how intrusive a system feature is perceived to be) may affect users' experience of focus, workflow, and sense of control in everyday work situations.

The results will be used in a thesis project within interaction design and may also be presented in academic contexts.

Your Participation

Participation consists of an individual semi-structured interview where you will be presented with short scenarios describing how different digital system features may be presented in a work environment.

During the interview, you will be asked to reflect on how these features are perceived and how they may affect your focus, workflow, and sense of control in your work.

The interview is expected to take approximately 45–60 minutes and can be conducted either digitally or in person.

With your consent, the interview will be audio recorded to facilitate analysis. The recording will only be used for research purposes and will be deleted after it has been transcribed (i.e., converted into written text).

Data Handling and Confidentiality

All information collected during the study will be treated confidentially.

Your name or any other identifying information will not appear in the thesis. All material will be anonymized during transcription.

The collected data will be used solely for this thesis project and stored securely in

accordance with applicable guidelines.

Voluntary Participation

Your participation in the study is entirely voluntary. You may withdraw at any time without providing any reason.

If you choose to withdraw, all material that can be linked to you will be deleted.

Contact

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* Indicates required question

1. **Consent** *

I have read the information above and agree to participate in the study.

Check all that apply.

☐ I agree

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Appendix 2: Interview Guide

Interview Guide

1. Background Questions

Can you briefly describe your role and the types of tasks you work with?

How would you describe your need for focus in your work? Are there situations in your work that require extra concentration?

What types of notifications, reminders, or system messages do you encounter in your work today? *(For example: email, chat functions, tasks in other systems)*

Core Questions per Scenario

First reaction
What is your immediate reaction to this situation?

Focus
How do you think this would affect your focus?

Workflow
How do you think this would affect your workflow or your ability to continue the task?

Perceived control
Would you feel that you are in control of the situation, or that the system takes over too much?

Supportive vs. disruptive
Would this feel helpful or disruptive? Why?

Timing
Are there situations where this would work better or worse?

Context
In what types of tasks would this be most and least appropriate?

3. Comparative Section

Which scenario felt the most disruptive?
Which worked best in real work situations?
What determines whether something feels reasonable or too intrusive?
Is there a middle ground that would have been better?

Appendix 3: Overview of AI tools used in this thesis

In this study, we used AI tools (ChatGPT, NotebookLM and Clipto AI) which were used selectively as supportive tools during parts of the research and writing process. The tools were primarily used for:

- Brainstorming and discussing ideas relevant to the research topic
- Supporting an overview of literature, articles, and theoretical concepts.
- Reflection and discussion of ideas, arguments and alternative perspectives
- Transcribing and summarising interview recordings
- Improving language, grammar and clarity
- Exploring alternative ways of presenting findings, figures and tables

All analysis, theoretical positioning, and conclusions presented in this thesis were developed by the authors. The final visual figures, tables, and mockups included in the thesis were created by the authors in Figma.

Appendix 4 Division of work Between writing partners

The work on this thesis was conducted collaboratively through the project. Both of us have contributed to the entire process, including planning, research design, development of scenarios, interviews, analysis, writing, and revision of the thesis. All parts of the work were continuously discussed and developed together.

Maria Alnaser

Tove Jakobsson